

I need to document steps and also all functions and approach for next time visit. can you give it to me for the final code I attached based on what you wrote for me?

Thought for 5s >

Yes. Below is a **clean documentation guide** for your final MATLAB calibration code, based on the attached script. It is written so you can keep it with the code and understand what each step, function, and fitting approach is doing next time.  finalcode

Documentation for final MATLAB calibration code

1. Purpose of the script

This script calibrates **multiple optodes** using the **modified Stern–Volmer model** from CSV calibration files. It does four main things:

1. Reads all calibration CSV files in the folder.
2. Computes the ratio

$$R = \frac{\text{Red}}{\text{Green}}$$

and propagates uncertainty from Red and Green standard deviations.

3. Fits each optode **independently**.
4. Performs a **joint weighted fit** across all optodes using:
 - shared α
 - shared K_{SV}
 - separate R_{high} for each optode

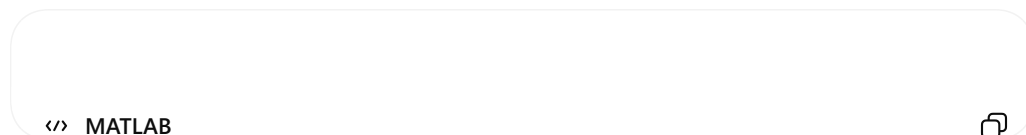
It then plots:

- calibration data,
- independent fits,
- joint shared-shape fits,

and saves a summary table.

2. Expected input format

The script automatically searches for files named:



Each CSV file must contain:

- **Column C:** mean intensity
- **Column D:** standard deviation of intensity
- **Column F:** oxygen concentration
- Rows arranged as:

- Green
- Red
- Green
- Red
- ...

So each Green/Red pair represents **one replicate** at one O₂ concentration.

3. Main workflow

Step 1. Find all calibration files

The script scans the current folder for all files matching:

<>
MATLAB

Then it:

- sorts them alphabetically,
- stores file names,
- creates plot labels from the file names.

This allows the script to work automatically for 2, 3, or more calibration files.

Step 2. Read each CSV file

For each file, the script reads:

- Mean from column C
- SD from column D
- C_a11 from column F

Then it removes invalid rows:

- NaN
- Inf
- missing values

If the number of rows is odd, the last row is removed so Green/Red pairing stays valid.

Step 3. Build replicate ratios

Because rows alternate Green and Red:

- odd rows are Green
- even rows are Red

The code computes:

$$R_{\text{rep}} = \frac{Red}{Green}$$

for each replicate pair.

Step 4. Propagate uncertainty for the ratio

The uncertainty of the ratio is computed using propagation of error:

$$\sigma_R = R \sqrt{\left(\frac{\sigma_{Red}}{Red}\right)^2 + \left(\frac{\sigma_{Green}}{Green}\right)^2}$$

This gives one uncertainty value for each replicate ratio.

Then invalid replicate values are removed:

- non-finite values
- zero or negative uncertainty
- zero/negative Green or Red intensity

Step 5. Group replicates by oxygen concentration

At the same O₂ concentration, you may have multiple replicate image pairs.

For each unique O₂ concentration, the script computes a **weighted mean ratio**:

$$\bar{R} = \frac{\sum w_i R_i}{\sum w_i} \quad \text{with} \quad w_i = \frac{1}{\sigma_{R,i}^2}$$

and its standard error:

$$\sigma_{\bar{R}} = \sqrt{\frac{1}{\sum w_i}}$$

This is better than a simple average because more precise replicates get more weight.

The output of this step is:

- one calibration ratio per O₂,
- one uncertainty per O₂.

4. Independent modified Stern–Volmer fit

Model used

For each optode, the script fits:

$$R(C) = R_0 \left[\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C} \right]$$

Parameters:

- α : non-quenchable fraction
- K_{SV} : Stern–Volmer constant
- R_0 : ratio at zero oxygen

The parameter vector is:

Weighted fitting

Each O₂ calibration point has an uncertainty R_{sd} , so the fit is weighted:

$$\text{objective} = \sum_i \frac{(R_i - R_{\text{model},i})^2}{\sigma_i^2}$$

This means:

- small uncertainty → large influence on fit
- large uncertainty → small influence on fit

The fit is solved with `fmincon`.

Confidence interval at measured datapoints

After fitting, the script evaluates the model only at the measured O₂ datapoints, not on a smooth grid.

It then estimates uncertainty of the fitted curve using:

1. numerical Jacobian J
2. weighted covariance matrix:

$$\text{Cov}_p = s^2 (J^T W J)^{-1}$$

3. propagated uncertainty in the fitted ratio:

$$\sigma_R^2 = J \text{Cov}_p J^T$$

A 95% confidence interval is computed as:

$$R \pm 1.96 \sigma_R$$

These are stored as:

- `Rlo`
 - `Rhi`
-

5. Joint weighted fit

This is the most important part if you want **one shared calibration behavior** across multiple optodes.

Why joint fitting is used

Different optodes may:

- have the same physical quenching response,
- but different absolute ratio baselines due to imaging/setup differences.

So the script assumes:

- α is shared

- K_{SV} is shared
- R_{high} is different for each optode

This is appropriate when:

- curves look similar,
- but are slightly shifted vertically,
- and you want one common calibration shape.

Joint model used

The script uses:

$$R(C) = R_{\text{high},f} \frac{\alpha + (1 - \alpha)/(1 + K_{SV}C)}{\alpha + (1 - \alpha)/(1 + K_{SV}C_{\text{high},f})}$$

Where:

- f = optode index
- $R_{\text{high},f}$ = ratio at the highest O_2 for optode f
- $C_{\text{high},f}$ = highest O_2 concentration in that calibration

So each optode has its own high- O_2 anchor, but all optodes share the same shape.

Why use R_{high} instead of R_0

This version is especially useful for your microfluidic experiments because:

- the new device often starts in a **fully oxic** state,
- but not necessarily a fully anoxic one,
- so the fully oxic image is a better reference.

That means:

- the shape (α, K_{SV}) can be transferred,
- the new optode gets its own high- O_2 reference from the experiment.

How joint fitting is done

The script stacks all calibration points from all optodes into one large dataset:

- all concentrations
- all ratios
- all uncertainties
- optode ID for each point

It then fits one parameter vector:

<> MATLAB
▢

with weighted least squares.

This gives:

- `alpha_joint`
 - `Ksv_joint`
 - `Rhigh_joint(f)` for each optode
-

6. Plotting approach

The final figure shows, for each optode:

- **colored circles** = weighted calibration data
- **vertical error bars on circles** = uncertainty of calibration points
- **white squares** = independent fit evaluated at datapoints
- **thin whiskers on squares** = 95% CI of independent fit
- **colored line** = joint fit evaluated at datapoints

So the figure allows you to compare:

- raw calibration data,
 - optode-specific fit,
 - shared-shape fit.
-

7. Output files

`calibration_summary.csv`

The script saves a summary table containing:

- optode label
- independent α
- independent K_{SV}
- independent R_0
- joint R_{high}

This gives a quick comparison between optodes and the pooled calibration.

8. Functions used in the script

Built-in MATLAB functions

`dir`

Used to find all calibration CSV files in the folder.

`readtable`

Reads each CSV file into a MATLAB table.

`isfinite`

Used to remove invalid numeric values.

`unique`

Finds the unique oxygen concentrations.

`sort`

Sorts oxygen concentrations and corresponding ratios.

`fmincon`

Performs constrained nonlinear optimization for both:

- independent fits
- joint fit

`optimoptions`

Defines the optimization settings for `fmincon`.

`arrayfun`

Used to extract fitted parameters from the `cal` structure.

`errorbar`

Plots data points and uncertainty bars.

`plot`

Plots individual fit points and joint fit curves.

`legend`, `xlabel`, `ylabel`, `title`, `grid`

Used to format the final plot.

`table`, `writetable`

Used to create and save the calibration summary file.

Local function

`local_joint_model_Rhigh`

This function defines the joint model used in the pooled fit.

Inputs:

- `p` : parameter vector `[alpha, Ksv, Rhigh_1, ..., Rhigh_n]`
- `c` : oxygen concentrations
- `C_high` : highest oxygen concentration for each datapoint's optode
- `id` : optode index for each datapoint

Output:

- predicted ratio R

This function is the core of the shared-shape fit.

9. Recommended usage next time

If you are calibrating new optodes

1. Export CSV files with the same format.

2. Put them in the same folder as the MATLAB script.

3. Name them like:

- results_opt1.csv
- results_opt2.csv
- results_opt3.csv

4. Run the script.

5. Check:

- console output
 - final plot
 - calibration_summary.csv
-

If you want one calibration behavior for a new microfluidic device

Use:

- alpha_joint
- Ksv_joint

Then determine the new device's:

- $R_{\text{high,new}}$

from the initial fully oxic image.

That is the most transferable and physically sensible approach.

10. Important assumptions and cautions

Assumptions

- Red and Green uncertainties are independent.
- Replicate errors are approximately Gaussian.
- The modified Stern–Volmer form is appropriate for the optode chemistry.
- Different optodes share the same shape in the joint fit.

Cautions

- If two optodes differ strongly in shape, do not force a joint fit.
 - If calibration images have drift, averaging will not fix systematic error.
 - If the highest O₂ point is unstable, R_{high} anchoring will be biased.
 - The confidence intervals here describe **calibration fit uncertainty**, not full experimental uncertainty.
-

11. Short interpretation guide

Independent fit

Tells you how each optode behaves on its own.

Joint fit

Tells you the best shared calibration behavior across all optodes.

If independent fits are close

The joint fit is meaningful and can be used as the common calibration shape.

If independent fits differ strongly

The optodes may not share the same Stern–Volmer response, and separate calibration may be safer.

12. Suggested header comment for your code

You can paste this at the top of your MATLAB file:

```
% This script calibrates multiple optodes using modified Stern–Volmer fitting.
% Input files must be named results_opt*.csv and contain:
%   column C = mean intensity
%   column D = standard deviation
%   column F = O2 concentration
% with rows alternating Green / Red.
%
% Workflow:
%   1. Read all calibration CSV files
%   2. Compute R = Red/Green
%   3. Propagate uncertainty for R
%   4. Group replicate image pairs by O2 using weighted mean
%   5. Fit each optode independently with modified Stern–Volmer model
%   6. Perform a joint weighted fit with shared alpha and Ksv,
%       and separate Rhigh for each optode
%   7. Plot data, independent fits, and joint fits
%   8. Save summary results to calibration_summary.csv
```

If you want, I can turn this into a polished **README-style document** or a **Word/PDF lab note** you can keep with the MATLAB code.

Sources



give me a word file

Your Word document is ready.

Download it here:

[Download the Word file](#)

It contains:

- purpose of the MATLAB calibration script
- input CSV structure
- step-by-step workflow
- explanation of the modified Stern–Volmer model